

IT & Cybersecurity Infrastructure Operations Manual

Sample Chapter (Public Excerpt)

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This document is a sanitized excerpt from a senior capstone engagement at Pensacola Christian College. Gulf Coast Family Dental, P.A. is a fictional client; specific environment details have been redacted from this public version. Full deliverables available on request.

About this excerpt

This is one section (7.7) excerpted from a 100+ page IT & Cybersecurity Infrastructure Operations Manual produced as part of a senior capstone engagement. The full manual is the standing reference document for the client's IT staff — covering environment architecture, server configuration, database documentation, security policies, software inventory, operational procedures, monitoring, maintenance, and troubleshooting.

Section 7.7 was selected for public excerpt because it demonstrates **analytical security writing** — identifying known gaps in an environment and recommending remediation — rather than step-by-step procedural documentation. Specific environment details (account names, hostnames, IP addresses) have been redacted from this public version.

7. Security Policies

7.7 Current Security Gaps & Recommendations

This section documents known security gaps identified during the implementation and assessment of the environment. Each item includes a description of the risk and a recommended remediation action. These items should be reviewed and addressed by authorized IT and security personnel on a priority basis.

7.7.1 Third-Party Consultant Accounts

Active Directory accounts were created for outside consultants to support the design and implementation of this environment. These accounts currently hold administrative privileges and remain active. Accounts with elevated privileges that are no longer needed represent an unnecessary attack surface and a potential compliance concern under least-privilege principles.

Recommendation

Upon formal project completion and handoff, all consultant accounts should be disabled immediately and flagged for deletion after a 30–90 day retention period, consistent with the account lifecycle policy documented in Section 9.4. Disable rather than delete initially to preserve auditability for any post-handoff investigation or compliance review.

7.7.2 Windows Activation Status

Endpoint workstations and server systems are currently running unactivated copies of Windows. Unactivated systems may be restricted from receiving certain updates and security patches, which could leave the environment exposed to known vulnerabilities over time.

Recommendation

Obtain and apply valid Windows licensing for all systems. Ensure all endpoints and servers are fully activated and enrolled in Windows Update to maintain patch compliance and continuous delivery of security updates.

7.7.3 Patient Portal Security

The patient portal is currently in development. As noted in Section 9.4, the portal is using **MD5** hashing for password storage. MD5 is a cryptographically broken algorithm and is not considered acceptable for storing credentials or sensitive data, particularly in a HIPAA-regulated environment.

Recommendation

Replace MD5 with a modern password hashing algorithm such as **bcrypt**, **Argon2**, or **PBKDF2** before the patient portal goes live. No patient credentials or protected health information should be processed through the portal until this is remediated.

7.7.4 Group Policy Object Description

Current Group Policy Objects lack descriptive naming and documentation, which makes auditing, troubleshooting, and change management more difficult. Without clear GPO descriptions, administrators may inadvertently duplicate or conflict with existing policy, and audit reviewers cannot quickly determine the intended purpose of each policy object.

Recommendation

Ensure all GPOs have clear, descriptive names and documented descriptions that reflect their purpose and scope. Document any GPO changes in the change management log maintained by IT administration, consistent with the change management procedures in Section 11.7.

End of sample excerpt. The full IT & Cybersecurity Infrastructure Operations Manual (100+ pages) covers environment architecture, server configuration, database documentation, security policies, software inventory, operational procedures, monitoring, maintenance, and troubleshooting. Available on request — contact Travis Currier, tcurrier1819@gmail.com.